



FIN 413 A01 Risk Management

Fall 2023 (5 Sep – 8 Dec 2023)

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Introduction

Separate from business risks, most corporations and financial institutions also face financial risks arising from changes in market variables external to their business such as commodity prices, interest rates, foreign currency exchange rates, and stock prices. Derivative contracts such as forwards, futures, options, and swaps have become important tools for hedging these financial risks. Due to the relatively low transaction costs and opportunities for speculative profits, the volume of derivatives traded often exceeds the quantity of the underlying assets being traded.

Risk management requires strong analytical and quantitative skills. There is a growing demand for individuals who are skilled at selecting and pricing derivatives. This course studies the characteristics of derivative contracts including forwards, futures, options, and swaps. The aims are:

- To know the characteristics of derivatives and their markets;
- To understand the theoretical approaches for determining their values;
- To be able to design and evaluate strategies for using derivatives to reduce financial risks.

Studying derivative securities inevitably becomes quantitative and technical. You will be dealing with a lot of math theory and computations. The challenge is in interpreting the numbers into notions of risk and value. We can then make practical choices to build desired risk management positions.

Learning Goals

Upon completion of this course, students should be able to demonstrate knowledge of derivative markets, mechanics of trading, financial payoffs, security pricing, and strategies for risk management.

This course incorporates the Learning Goals of the Alberta BCom Program, in particular Critical Thinking, Written Communication, and Quantitative Skills.

Textbook and Class Materials

- Textbook: *Either* Hull. Fundamentals of Futures and Options Markets, 9th Edition, 2016.
 Or Hull. Options, Futures, and other Derivatives, 10th Edition, 2018.
 The new online-only 10th Edition of Hull Fundamentals may also be okay.
- Class materials: Class materials such as lecture slides, assignment information, and selected additional readings will be placed on eClass (<https://eclass.srv.ualberta.ca>).
- Additional reading: Solutions and study guide for the textbook. Newspapers such as the National Post, Globe and Mail, Wall Street Journal, and online news feeds often mention derivative markets. Any articles discussed in class may become exam material.

Readings and Class Schedule

The proposed class schedule is as follows. The schedule may be modified as the semester progresses if topics take more or less time than planned. Readings and/or chapters from the text may be added or deleted at a later date. Please ensure you have read the relevant materials before each session.

Date	Topic	Readings Hull Fund 9e	Readings Hull Opt 10e/11e
Sep 5 Sep 7	Introduction. Futures markets. Energy futures.	Chs. 1 and 2 Ch 24.2	Chs. 1 and 2 Ch 35.3
Sep 12 Sep 14	Futures hedging strategies. Forward and futures prices.	Ch 3 Ch 5	Ch 3 Ch 5
Sep 19 Sep 21	(continued) Interest rates.	Ch 4	Ch 4
Sep 26 Sep 28	Interest rate futures. Review	Ch 6	Ch 6
Oct 3 Oct 5	First Midterm Exam (online during class) Mechanics of options. Bounds on option pricing.	Chs. 9, 10, 11	Chs. 10, 11, 12
Oct 10 Oct 12	Options trading strategies. Binomial option pricing.	Ch. 12	Ch. 13
Oct 17 Oct 19	(continued) Black-Scholes option pricing.	Chs. 13 & 15	Chs. 15 & 17
Oct 24 Oct 26	Options on indices and forex. Futures Options and Black's Model. Review.	Ch. 16	Ch. 18
Oct 31 Nov 2	Second Midterm Exam (online during class) The Greek Letters.	Ch. 17	Ch. 19
Nov 7 Nov 9	Delta. Theta. Gamma. Vega. Rho. Higher order. Implied Volatility. Volatility smiles.	Ch 19.	Ch. 20
Nov 13	Reading week – no classes	n/a	n/a
Nov 21 Nov 23	(continued) Teamwork assignment due. Value at risk (VaR) and Expected Shortfall (ES).	Ch. 20	Ch. 22
Nov 28 Nov 30	Swaps. Securitization and the credit crisis.	Ch. 7 Ch. 8	Ch. 7 Ch. 8
Dec 5 Dec 7	Credit derivatives. Exotics. Derivative mishaps. Review.	Ch. 23 Chs. 22, 25.	Ch. 25 Chs. 26, 37.
	Study and exam week – no classes	n/a	
Dec 12	Final Exam – check BearTracks for date & venue. expected to be Tue Dec 12 at 9:00am	All the above	All the above

The lecture venue is BUS 3-10 unless otherwise advised through BearTracks or eClass.

Grading

Course grades will be determined by combining points on participation, homework assignments, two midterm exams, and a final exam. The following table shows the component marks available to students. Grades will be allocated after considering the relative performance of each student.

Participation	5% or 0%	This 5% will be awarded as the higher of your participation and your final exam. Participation will be assessed throughout the course and evaluates critical thinking and verbal communication skills as well as problem solving - not merely attendance. If you are unable to participate, or if your final exam score is higher than your participation mark, this 5% will be redistributed to your final exam.
Online practice	5%	Online practice is provided at https://GradeChaser.com . Please register using your ualberta.ca email address (if you have had more than one uAlberta email address please ensure you use the one that is currently registered on eClass.) Your scores will be recorded three times corresponding to the midterm and final exams. The snapshot will be taken during or after each exam so that you can also use GradeChaser.com during your revision and continue to earn marks. You should complete the new topics for each exam in time for that exam. In the event that you are granted a deferred exam, you may also request that your deadline for the online practice be extended to match the deferred exam.
Assignment	10%	The teamwork assignment involves problem solving, teamwork, leadership, quantitative analysis, information processing, and written communication. Each group should make one submission. The team size can be 3, 4, or 5 students. The teamwork assignment will be available on the course website at least two weeks prior to the deadline. Please submit your files through eClass by the deadline, noting that late submissions may receive zero without being marked.
Midterm exams (60 minutes x 2)	20%+ 20%	These assess individual critical thinking and problem solving as well as quantitative and information processing skills, and the grasp of material. The midterm exams will be online and open-book, with further details below.
Final exam (120 minutes)	40% or 45%	This assesses critical thinking and problem solving as well as quantitative and information processing skills, and the grasp of the material. The final exam covers the whole course, although with more emphasis on the portions not covered by the midterm exams. The final exam will be in-person and closed-book, with further details below.
TOTAL	100%	

If your online midterm marks are significantly higher than your final exam, or if there are any anomalies with your midterm exams, I may ask you to rewrite closed-book paper midterms to demonstrate that you know the material, otherwise the midterm weights may be redistributed to the final exam.

There may be a requirement to pass the final exam in order to pass the course.

In addition to the graded work, there will also be further end-of-chapter chapter questions and exercises without credit for you to work on throughout the term to gain a greater understanding of the material.

Online Practice

The main purpose of the online practice at <https://GradeChaser.com> is to help you understand the material. You can attempt each question up to ten times and I provide detailed worked solutions.

Please ensure that you select the correct course. Click

Change Course

Select your course:

University of Alberta

Introduction to Finance FIN 301

Risk Management FIN 413

Fixed Income FIN 418 and 618

You are welcome to explore the other courses too if you wish, but make sure you switch back.

You earn a mark for each question if at least half of your attempts are correct. You can attempt each question up to ten times. Once you have earned the mark for a question you will keep the mark even if you subsequently make more attempts for practice and get them wrong. You know that you have earned the mark for a question when you see a smile alongside the question. The symbols are:

	Question not yet attempted.		Mark earned on first attempt.
	Mark not yet earned. More than half the attempts so far have been incorrect.		Mark earned (at least half of your attempts were correct).
			Mark retained (you keep your mark once it is earned even if subsequent additional attempts are wrong).

If you have scored the mark for at least 80% of the GradeChaser questions you will be given the full 100% for the online practice component. This means that you do not need to perfect every question to get full marks. For example if you miss out on the mark for a question because of some minor error and you are satisfied that you know what to do, you can move on without having to redo it, and still get full marks provided you have completed at least 80% of the questions correctly.

You can tell your overall score for the course by clicking on the whole course at the top and then reading the score on the bottom of the screen:

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You have scored 1.9% of these **GradeChaser** practice problems.

Please ensure you register using the same ualberta.ca address as in eClass if you have had multiple ualberta addresses. Your grade will be matched using the eClass email address.

If you already have some scores from using GradeChaser previously, you can either keep those scores or I can reset your account to zero so you can do it afresh. Contact me if you would like it reset.

Further Practice

In addition to the graded work, there are also end-of-chapter problems and exercises for you to work on throughout the term to gain a greater understanding of the material, without credit. The solutions will be placed on eClass.

Sample representative exam questions will be discussed in class prior to the examinations. Many past exam questions with randomization are available at <https://GradeChaser.com>.

Midterm Examinations

The midterm examinations will be held online at my web site <https://KeithGodfrey.net> and they will be open book and open-computer. You can access any textbooks both physically and online, including the course materials on eClass, and you can even use Google to look up anything that you like - but you must not communicate about the exam materials with anyone (except for your instructor) or any tool (such as ChatGPT) until after the exam time window has finished for everyone.

To clarify: You are welcome to look up reference material anywhere, but it is considered cheating to send any part of the exam (or similar concept) to any place that could help produce an answer. Examples of what is acceptable and not acceptable:

- | | |
|----------|---|
| Okay | <ul style="list-style-type: none">• Looking up a word (e.g. fortnightly) in an online dictionary.• Looking up a term (e.g. Black Scholes) in Google, Wikipedia, Investopedia, etc. |
| Not okay | <ul style="list-style-type: none">x Sending a question (or asking something related) to someone else, maybe a friend, maybe an online tutoring team like Chegg or Course Hero, so that they can work towards an answer or explanation.x Sending a question (or asking something related) to an Artificial Intelligence tool such as ChatGPT or Bard so that it can develop an answer or explanation. |

The rule is that you must do the work yourself to develop your answers. That is what you are graded on. It is cheating when someone else or something else is doing any part of that work.

It is also cheating to use any answer that has come from somewhere other than yourself (e.g. if a friend sends you an answer regardless of how they obtained it).

Unauthorized communication by any means will be prosecuted to the maximum extent of the university policy, including using materials arising from other people's unauthorized communications. If you happen to find the questions or answers somewhere, it likely means someone is trying to cheat – please contact me immediately so this can be resolved. You should not be penalized if you declare this immediately.

The questions in our on-line exams will be designed so that the answers are not available through Google or in the lecture notes. Online questions will typically be more numerical than theory and will require demonstration or application of theory. You need to know something about the material. The grading is based on your demonstrated knowledge.

You will need access to a PC / laptop / notebook with software including a modern web browser (such as Chrome) and Microsoft Excel or similar spreadsheet package. You will not be expected to know the advanced features of Excel – this is not a test of Excel – but you should know the basics of how to read and edit a spreadsheet if required.

Any student requiring special accommodations should contact me at least two weeks before each exam.

Final Examination

The final examination will be a traditional closed-book paper exam scheduled by the university.

You may bring pencils, pens, calculator(s), and transparent bottled water. You must display your ONECard identification on your desk.

Closed book means that no books or notes of any kind are allowed. A formula sheet will be provided to you in the examination. The formula sheet will be discussed in class and shown in advance on eClass (but do not bring your own copy – it will be provided to you).

Computers, notebooks, laptops, tablets, and any devices with communication facilities are not allowed. Programmable calculators must be "de-programmed" before exams. You can bring multiple calculators (for example both a financial calculator and a scientific calculator) provided that each calculator is deprogrammed. If you want to retain programmed information then you should bring a different calculator to the exams. Ensure that you know how to drive your calculator(s) and that your batteries have sufficient power. A financial calculator can be useful but is not essential. A calculator that has y^x , e^x , and $\ln(x)$ functions should be sufficient.

Missed Assessment and Deferred Exams

The rules for attendance and absence from term work and examinations are specified in the Academic Regulations (<http://calendar.ualberta.ca/content.php?catoid=6&navoid=806>). If you miss, or if you know you are going to miss any term work or exam for a reason that is acceptable to the university (such as an incapacitating illness, a severe domestic affliction, a religious belief, and so on) then please provide supporting evidence as soon as possible. You will receive a mark of zero in a missed assessment unless appropriate supporting documentation is received in a timely manner.

If you miss the **online practice** or **teamwork assignment** for a reason that is acceptable to the university, please supply supporting documentation to me within two days (or as soon as practical in cases of extenuating circumstances). With evidence provided in time, the deadline for the online practice can be changed, or the weight from the missed assignment may be redistributed to the other components.

If you miss a **midterm exam** for a reason that is acceptable to the university, please supply supporting documentation to me within two days (or as soon as practical in cases of extenuating circumstances). With evidence provided in time, the weight from the missed exam may be redistributed to the other exams.

If you miss the **final exam** for a reason that is acceptable to the university, please contact your faculty office promptly in accordance with the University Regulations and Information for Students in the University Calendar. The office will decide whether to award a deferred final exam.

The deferred final will be held on **Wednesday 14 February 2024 at 1:30pm** unless an alternative arrangement is agreed.

Disagreements with Assessments

Your grades will stand as posted online unless any mistakes are drawn to my attention in a timely manner. Grades will normally be available within one business week of exams. Once a final grade is assigned, it will not be changed except in the case of a recording error. If you feel that your midterm mark or final mark is incorrect, please notify me by email within one week from the posting of the mark online (or as soon as practical in cases of extenuating circumstances). Please note that any query about an exam mark will normally result in the entire exam being remarked which may lead to your mark going up or down. I may send your paper for remarking by a different marker, or I may invite you to witness me remarking the paper in cases where your perception of your mark is substantially different from mine.

Office Hours and Email

My office hours will initially be Thursdays 11:00 to 12:00 in room 1.23(b) and will initially be shared with all my courses. Any changes and additional times will be posted on eClass. I am also usually available after each class and otherwise by appointment.

My email addresses are keith1@ualberta.ca for course matters and admin@gradechaser.com for any issues with the online practice. I normally respond to emails within two business days or usually faster. In the subject line please include the course, and please include your name in the email.

Please use your university email address. Emails sent to any ualberta address from non-university accounts risk being rejected by the university's spam filter and might never be received or read.

University of Alberta Rules

Policy about course outlines can be found in Course Requirements, Evaluation Procedures and Grading of the University Calendar.

The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Code of Student Behaviour (online at www.governance.ualberta.ca) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts and/or participation in an offence. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study, and is not to be used or distributed for any other purpose without prior written consent from the content author(s).